

PLYMOUTH, NH, USA

WMO: 742078

Lat: 43.779N

Lon: 71.754W

Elev: 505

StdP: 14.43

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 64773

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.7	0.0	-14.9	2.5	-2.0	-10.5	3.2	4.0	22.0	19.6	20.1	18.4	5.0	280	0.505

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	87.7	70.1	84.4	68.8	81.8	67.1	73.3	82.3	71.6	80.5	69.9	77.7	5.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.5	114.2	77.2	68.8	107.8	75.5	67.5	102.8	74.0	37.2	82.5	35.7	80.7	34.2	78.1	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	16.3	13.4	-14.5	92.3	5.8	1.7	-18.7	93.5	-22.1	94.6	-25.3	95.6	-29.5	96.8
			-14.7	76.3	5.5	1.7	-18.7	77.5	-21.9	78.5	-25.0	79.4	-28.9	80.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.4	20.4	22.7	31.1	43.4	55.8	63.3	68.6	66.7	59.7	48.8	36.4	26.7
	DBStd	18.66	10.42	9.98	9.66	8.11	7.49	6.15	5.09	4.89	7.49	7.61	7.91	9.15
	HDD50	3783	917	765	590	230	25	0	0	0	7	116	412	721
	HDD65	7470	1382	1185	1052	648	302	103	23	38	193	501	857	1186
	CDD50	2117	0	0	4	32	204	400	576	518	298	80	5	0
	CDD65	330	0	0	0	1	15	53	134	91	35	1	0	0
	CDH74	3588	0	0	8	40	327	644	1352	848	349	20	0	0
	CDH80	974	0	0	0	10	91	179	413	188	92	1	0	0
(14) Wind	WSAvg	3.9	5.1	5.4	5.4	4.9	3.8	3.2	2.5	2.2	2.4	3.4	3.9	4.7
(15) Precipitation	PrecAvg	43.7	3.0	2.6	3.5	3.4	3.4	4.0	4.4	3.9	3.8	4.6	3.7	3.8
	PrecMax	54.6	5.8	6.5	5.3	6.5	8.1	8.8	7.9	9.4	8.6	10.8	5.5	7.0
	PrecMin	30.6	0.7	1.1	1.7	0.6	0.5	1.3	1.7	0.7	1.5	1.1	2.0	1.6
	PrecStd	6.4	1.3	1.1	1.0	1.5	1.8	2.0	1.5	2.0	1.5	2.8	0.9	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.6	53.3	68.3	79.5	88.0	90.4	91.8	88.5	87.9	77.1	64.2	51.4
		MCWB	40.1	42.9	55.2	59.6	67.8	69.9	73.5	71.9	71.7	62.2	56.0	48.4
	2%	DB	41.0	45.1	55.3	71.9	82.1	85.3	87.8	84.5	82.3	70.3	58.6	45.3
		MCWB	36.7	38.1	44.0	54.8	63.5	67.8	71.0	69.6	68.8	60.0	51.5	41.3
	5%	DB	37.0	39.1	48.8	65.7	77.2	80.9	84.5	81.9	77.9	65.1	53.8	41.8
		MCWB	33.7	34.5	40.9	51.5	61.2	64.8	69.6	68.1	66.1	58.2	48.8	38.5
	10%	DB	34.6	35.7	44.0	59.5	72.1	77.4	81.6	79.1	73.2	61.2	49.3	38.1
		MCWB	32.3	32.1	37.5	47.6	57.9	63.2	68.1	67.1	64.1	55.3	44.8	35.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.8	45.1	55.9	63.0	70.5	72.8	76.7	74.2	74.0	67.1	58.7	47.5
		MCDB	45.6	50.4	67.4	73.5	80.7	82.4	86.5	83.9	83.3	71.3	62.3	49.1
	2%	WB	37.1	38.5	47.0	58.5	67.3	70.8	74.1	72.2	71.0	62.3	53.6	42.5
		MCDB	39.7	42.9	52.5	69.8	77.2	80.9	83.1	81.0	79.9	67.3	56.9	44.2
	5%	WB	34.6	35.2	41.6	53.3	64.0	68.3	72.2	70.4	68.5	59.3	48.9	38.9
		MCDB	36.7	38.7	46.0	61.6	71.9	76.1	80.4	78.3	74.9	63.5	52.9	41.7
	10%	WB	32.5	32.8	38.4	49.8	61.1	66.1	70.3	68.9	66.0	56.6	45.5	35.9
		MCDB	34.2	34.9	43.4	58.7	68.5	73.5	77.8	75.7	71.4	60.8	48.2	37.9
(35) Mean Daily Temperature Range	MDBR		17.2	18.5	18.8	21.0	24.0	21.9	22.4	22.5	22.8	19.5	16.6	13.8
	5% DB	MCDBR	17.0	21.6	26.8	34.9	33.3	28.9	27.7	26.5	27.1	25.7	24.8	16.6
		MCWBR	14.1	15.9	16.9	18.5	16.3	13.3	13.2	13.4	14.6	16.0	17.9	13.8
	5% WB	MCDBR	15.5	19.6	23.4	28.6	26.1	23.5	23.1	23.0	23.4	20.9	21.5	15.4
		MCWBR	13.3	15.1	16.2	17.6	14.1	12.6	12.0	12.3	13.7	14.7	17.3	13.6
(40) Clear-Sky Solar Irradiance	taub		0.275	0.287	0.316	0.352	0.380	0.428	0.445	0.417	0.370	0.335	0.309	0.281
	taud		2.404	2.345	2.389	2.402	2.373	2.248	2.212	2.277	2.395	2.504	2.519	2.472
	Ebn at Noon		273	289	293	290	283	268	262	266	271	269	259	261
	Edh at Noon		24	30	33	36	38	43	44	40	33	26	22	20
(44) All-Sky Solar Radiation	RadAvg		503	779	1137	1412	1582	1712	1773	1609	1308	807	532	390
	RadStd		44	70	73	164	188	156	128	89	86	83	58	38

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (61 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	N/A

Nomenclature: See separate page